

Waves and wave dispersion for geophysical fluids

This module is part of a larger course dealing with different wave types and instability mechanisms in atmosphere and ocean dynamics. Here, we deal with some of the basic wave types and with general properties of waves, such as dispersion. We focus mainly on understanding the different wave types in isolation and therefore often make simplifying assumptions that are rarely valid in realistic settings. To understand more realistic flows, one must also consider how these different wave types and instability mechanisms interact with one another and with the general circulation.

Waves and instabilities as discussed here are both *linear* dynamical modes, solutions to the underlying PDE when linearised. This has a couple of important consequences: i) the behaviour of these solutions are amplitude-independent, so making them twice as big doesn't change their propagation or growth rate at all and ii) they obey a *superposition* principle, which means that the sum of two modes evolves just as if the individual modes evolved in time and were then added together, motivating us to choose a good "basis" in the sense of linear algebra. Further, we are discussing waves, not instabilities: these are modes that have a fixed "size" (energy, say) as they evolve in time, rather than growing or decaying. This requires waves to be associated with a physical *restoring* mechanism, that keeps dragging the perturbations back towards the mean state.

There are many adjectives and other modifiers used to describe waves relevant to oceanography and atmospheric science. Examples include gravity, inertia-gravity, surface, internal, equatorial, mountain, coastal, linear, nonlinear, solitary, Rossby (for which the restoring mechanism relates to the Coriolis parameter varying with latitude), Kelvin (for which the coast or equator acts as a waveguide), and so on. We will only consider a few here, but will develop techniques and background knowledge that are quite general and can be used to study other wave types later. Interesting problems arise as these different wave types interact with one another and with the background flow which is likely turbulent. Given this, one might ask whether it makes sense to bother studying linear waves in isolation as we intend to do here. A couple of reasons for doing so are that i) it is much simpler to first understand waves and wave propagation for different wave types individually and ii) this gives insight for understanding more complex flows and geometries than the ones we consider here.

We will consider propagation of linear waves through a uniform medium, ignoring boundaries and instead assuming the domain to be either periodic or infinite. This allows us to make use of Fourier's theorem, which gives that any signal of interest can be written as a sum or integral of sinusoids (sum if periodic and integral if infinite). By a uniform medium, we mean that the underlying equations don't have any parameters that vary in space or time, at least over some region and time frame of interest. These assumptions imply that the individual Fourier components comprising a more complex disturbance propagate independently of one another (in the language of linear algebra, they become *eigenmodes* of the system, and are in some sense an optimal basis, which they would not be for a different

domain geometry, or with a non-uniform medium). Given this, if we know how a generic sinusoidal wave propagates—and we have a Fourier representation of the signal at some initial time—then it is straightforward to reconstruct the total signal at later times. This motivates us to reason in Fourier space; because this often comes across as abstract, we will put a lot of focus on going back and forth between the two descriptions, i.e., to try and develop a better intuition for how the Fourier or spectral description translates to the real world. Thus this module is also a *practical* introduction to applied Fourier analysis, and to some tools from ‘`scipy.fftpack`’.

What will we be reasoning out in Fourier space? The main object of our analysis is a phenomenon called dispersion. If we leverage Fourier’s theorem, then we are explicitly writing any initial condition—or ‘signal’—as a sum of independent modes. If these modes propagate at different speeds, or in different directions, the signal, seen as an interference pattern of these modes, is going to change in shape over time. It is this phenomenon that we term “dispersion”, and the main goal here is that you understand it, both formally and intuitively.

Chapter One: The wave equation

A good starting point for a discussion of waves is ‘the’ wave equation. Written in one dimension,

$$\eta_{tt} - c^2 \eta_{xx} = 0,$$

where subscripts denote partial differentiation. You have likely seen this in various contexts. For example, for the case of a guitar string, c^2 works out to be given by the ratio of tension in the string and linear mass density. More relevant to fluids is the wave equation as implied by the linear shallow water equations. In this case, η is height above some appropriate zero and the parameters g and H correspond to gravity and layer thickness. Taking u and v to be the two horizontal velocity components, the non-rotating linear shallow water equations can be written as

$$\begin{aligned} u_t &= -g\eta_x \\ v_t &= -g\eta_y \\ \eta_t &= -H\delta, \end{aligned} \tag{1}$$

where $\delta = u_x + v_y$ is the horizontal divergence. Taking the time derivative of the mass equation and the divergence of the momentum equation leads to $\eta_{tt} = -H\delta_t = gH\nabla^2\eta$. Identifying $c^2 = gH$, this gives a 2-d version ($\eta_{tt} - c^2\nabla^2\eta$) of the 1-d wave equation written above.

Acoustic waves give another example of the wave equation in fluid mechanics. These, of course, are filtered by the incompressibility approximation. To see the connection with the wave equation, consider linearized momentum and mass equations in 1-d. We assume density perturbations to be small to a base state value, ρ_0 . This allows us to neglect

advection of density ($\mathbf{u} \cdot \nabla \rho$) with respect to $\rho_0(\nabla \cdot \mathbf{u})$. If, however, the time scale over which density changes is small compared to the advective time scale (L/U , where L and U are typical length and velocity scales), then we cannot necessarily discard the tendency term, i.e., ρ_t . Thus, in 1-d, we have

$$\begin{aligned}\rho_0 u_t &= -p_x \\ \rho_t &= -\rho_0 u_x \\ \rho &= \rho(p).\end{aligned}\tag{2}$$

The third equation above, is an equation of state in which we have simplified by considering the medium to have a uniform potential temperature (moisture and/or salinity is also considered uniform). Defining c by $c^{-2} = \partial_p \rho$, the second and third equations can be combined to give $p_t = -c^2 \rho_0 u_x$, from which the wave equation follows. [Meteorologists may want to use the definition of c above and the dry ideal gas law to derive an expression for the speed of sound in terms of temperature in Kelvin. In doing this, it is important to recall that it is potential temperature, not temperature, that is considered constant.]

An interesting way to solve the 1-d wave equation is to define an operator as follows,

$$\begin{aligned}\mathcal{L} &= \partial_{tt} - c^2 \partial_{xx} \\ &= \mathcal{L}_1 \mathcal{L}_2 \\ &= (\partial_t - c \partial_x)(\partial_t + c \partial_x),\end{aligned}\tag{3}$$

where $\mathcal{L}_1 = \partial_t - c \partial_x$ and $\mathcal{L}_2 = \partial_t + c \partial_x$. Clearly, solutions to $\mathcal{L}_1 \eta = 0$ or to $\mathcal{L}_2 \eta = 0$ are also solutions to the wave equation. It works out that $f_1(x + ct)$ is a solution to $\mathcal{L}_1 \eta = 0$. Similarly, $f_2(x - ct)$ is a solution to $\mathcal{L}_2 \eta = 0$. For example, define $\xi = x + ct$, and note that $\partial_x \xi = 1$ and $\partial_t \xi = c$. Then, $\mathcal{L}_1 \eta = \partial_t \eta - c \partial_x \eta = (c - c)df_1/d\xi = 0$. This solution, then, has some arbitrary shape given by $f_1(x)$ at $t = 0$ and propagates to the left at speed c . Similarly, a solution to $\mathcal{L}_2 \eta = 0$ has some shape defined by $f_2(x)$ at $t = 0$, and propagates to the right. For example, define $x_{\text{ref}} = ct$. Then, $f_2(x - ct) = f_2(x - x_{\text{ref}})$. Since $-x_{\text{ref}}$ becomes more negative with time, the x position of a particular feature in the signal (e.g., a local maximum, say) must increase with time. That is, the signal propagates to the right (assuming $c > 0$). This is also evident from inspection of the operator $\mathcal{L}_2 = \partial_t + c \partial_x$. It looks analogous to the Lagrangian derivative assuming a velocity field given by a uniform propagation in the positive x direction, e.g., $\partial_t + U \partial_x$ if $\mathbf{u} = U \hat{\mathbf{x}}$. The difference here is that the signal is not being advected to the right, it is propagating. Put another way, as the signal moves at speed c , the underlying water does not.

Now proceed to the Jupyter notebook for Exercise One, where we investigate: i) investigate the nondispersive propagation of sinusoids governed by “the” wave equation and ii) introduce dispersion in its simplest possible form.

Chapter Two: Fourier Decomposition

We have seen how sinusoids propagate in 1D, and we have seen how if they oscillate at different speeds, that gives rise to dispersion. But suppose now we have some arbitrary signal $f(x)$. If we know how to write it as the sum of many sinusoids

$$f(x) = \sum_k A_k \cos(kx) + B_k \sin(kx)$$

and further, if we know how to calculate $\omega(k)$, we can propagate each Fourier component (each sinusoid) and then add them back up at a later time, obtaining a solution. In this section, we deal with the first of these two techniques: decomposing the signal into sinusoids, or *Fourier analysis*.

This abstract technique works for a startlingly large set of periodic functions, as well as for localised functions defined on the (infinite) real line. In practice, it is inconvenient numerically to assume an infinite domain. Instead, we often assume periodicity, even for problems where what we really have in mind is a signal in infinite space. This is merely a convenience. Suppose, for example, that $f(x) \rightarrow 0$ as $|x| \rightarrow \delta L$, where δ is small. In words, suppose that the signal width is small compared to L . We might then take L as a stand-in for infinity, and it then make sense to represent f by its periodic analog, e.g. with $f(x + nL) = f(x)$ for all integers, n . Whereas the original signal was confined to a region near the origin, say, the periodic version also includes an infinite series of copies. Waves tend to propagate, and part or all of a signal originally confined to a region near $x = 0$ might later be farther away. Since this is also true for the infinite copies of the original signal, eventually, part of the solution in a original area of interest (e.g., near the origin) might have come from one of the copies. Moreover, portions of the disturbance originally near $x = 0$ may still be nearby. From this point forward in time, our analog of L being a stand-in for infinity has broken down. To continue using the periodic version of f as representative of an isolated signal on an infinite plane, we would have to use a larger value of L . If the signal propagates away, then eventually information from these copies propagates into our area of interest. But if we restrict attention to, say, the region between $x = -L/2$ and $x = L/2$ then for some finite time the solution behaves as it would in infinite space. How long depends on how fast the signal propagates and on the size of L . In other words, L is a large but arbitrary length scale over which the solution is assumed to be periodic, and how large is ‘large enough’ depends on the problem we wish to study.

The coefficient of x inside the trigonometric function is the *wavenumber*, and it represents the number of wave crests (or troughs) that fit in a period of length 2π ; it is thus defined by the equation $k = 2\pi/\lambda$. Consequently, the maximum wavenumber we can consider — called the Nyquist wavenumber — is given by $k = 2\pi/2\Delta x = \pi/\Delta x$ as that is the smallest resolvable wavelength. Similarly, the minimum nonzero k is simply $2\pi/L$. Since k takes values in an arithmetic progression, we can write $\Delta k = 2\pi/L$.

For the 1-d wave equation being considered, a Fourier presentation is not really needed, but is useful to serve as an example to be expanded upon later. Let us take $f(x)$ to be periodic and well represented by

$$f \approx \bar{f} + \sum_1^N A_n \cos(k_n x) + \sum_1^N B_n \sin(k_n x),$$

where the approximation is due to N being finite. Notation above is such that $k_n = 2\pi n/L$ and A_n and B_n are the sine and cosine coefficients. The first term is just $\bar{f} = A_0$ and corresponds to the average value over the domain. In practice, we would calculate these coefficients using an algorithm called the FFT (fast Fourier transform), but it is instructive to first consider how we might do so without one. Recall first that for any k_n, k_m pair, the average of $\sin(k_n x) \cos(k_m x)$ over an integer multiple of the two wavelengths is zero. This is also true for $\cos(k_n x) \cos(k_m x)$ and for $\sin(k_n x) \sin(k_m x)$ provided $n \neq m$. Recall also from basic calculus that the average of $\cos^2(kx)$ or $\sin^2(kx)$ over a wavelength is equal to $1/2$. Given this, to calculate A_m , say, we evaluate

$$C = (1/L) \int_{-L/2}^{L/2} [\cos(k_m x) f(x)] dx.$$

For example, this might be done numerically. Then, using the Fourier representation of f_1 , we have

$$\begin{aligned} C &= \overline{(\cos(k_m x) f_1(x))} \\ &= \overline{(\cos(k_m x) \sum A_n \cos(k_n x))} + \overline{(\cos(k_m x) \sum B_n \sin(k_n x))} \\ &= A_m \overline{\cos^2(k_m x)} \\ &= (1/2) A_m, \end{aligned} \tag{4}$$

where the overlines denote an average over L . In other words, A_m is just twice the average over L of $\cos(k_m x) f_1(x)$. A fast Fourier transform uses an evenly spaced grid and recurrence relations relating higher frequencies to lower frequencies, so as to calculate such averages efficiently. What is described above amounts to what I call a slow Fourier transform. It is inefficient—which becomes a problem when N is large—but is useful for illustrative purposes.

You are now ready to try Exercise 2.1.

Equivalent representations

As we hinted at (when we used complex exponentials) in the first notebook, there are many equivalent ways to carry out a Fourier transform.

As above, we take $f(x + nL) = f(x)$ for all integers, n . We can thus write

$$f(x) \approx \sum_0^N A_n \cos(k_n x) + \sum_0^N B_n \sin(k_n x).$$

A caveat is that $\cos(k_0 x) = 1$ for $n = 0$. Conversely, $\sin(k_0 x) = 0$ for all x . So, the coefficient B_0 is arbitrary and can be discarded. The coefficient A_0 is just the mean of $f(x)$. This can be rewritten as

$$f(x) \approx \sum_0^N C_n \cos(k_n x - \phi_n),$$

where $C_n = \sqrt{A_n^2 + B_n^2}$ and ϕ_n can be found from $A_n = C_n \cos(\phi_n)$, $B_n = C_n \sin(\phi_n)$. Similarly, we can also write

$$f(x) \approx \bar{f} + \sum_1^N C_n \sin(k_n x - \phi_n).$$

In this case, the mean is added explicitly and the ϕ_n obey $A_n = -C_n \sin(\phi_n)$ and $B_n = C_n \cos(\phi_n)$.

Using complex variables

Most useful is to move to complex notation. Define $\mathcal{A}_n = A_n + iB_n$. Equivalently, $\mathcal{A}_n = C_n e^{+i\phi_n}$. The complex coefficient encodes both the amplitude $|\mathcal{A}_n| = C_n$ and the phase of the wave! Then, using $e^{ik_n x} = \cos k_n x + i \sin(k_n x)$, we can write $f(x)$ as

$$f(x) = \text{Re} \left(\sum \mathcal{A}_n e^{ik_n x} \right)$$

It is straightforward to see that the derivative $f'(x)$ becomes

$$f'(x) = \text{Re} \left(\sum (ik_n) \mathcal{A}_n e^{ik_n x} \right).$$

Taking higher order derivatives is similarly easy, for example

$$f''(x) = \text{Re} \left(\sum (-k_n^2) \mathcal{A}_n e^{ik_n x} \right),$$

and so on.

In all of this, we have assumed that we are taking the real part of a sum of complex exponentials. But in full generality, a Fourier transform is defined for complex-valued functions as well. That is a complex function $f(x) : \mathbf{R} \rightarrow \mathbf{C}$ can also be written as a sum of complex exponentials —no real projection needed.

Wait a minute. A real-valued function is just a subtype of the complex one. That implies that there should be a way to write a *real* field directly as a sum of complex exponentials as well. We can do this by including, for every e^{ikx} , a Fourier component for e^{-ikx} . We want the imaginary component to cancel out for all x . Now this requires:

$$\begin{aligned} \text{Im}[\mathcal{A}_n e^{ikx} + \mathcal{A}_{-n} e^{-ikx}] &= 0 \\ \implies A_n \sin(kx) - B_n \cos(kx) + A_{-n} \sin(-kx) - B_{-n} \cos(-kx) \end{aligned}$$

And this is satisfied if

$$\mathcal{A}_n = A_n + iB_n \quad \mathcal{A}_{-n} = A_n - iB_n$$

In other words, the Fourier component at $-n$ is the complex conjugate of that at $+n$. You may recognise this from complex algebra as the *Hermitian property*. We have cancelled out the imaginary part, but in doing so have doubled the real part.

Suppose we carry out the real Fourier transform to get the coefficients $\{\mathcal{A}_0, \mathcal{A}_1, \dots, \mathcal{A}_n\}$. To get from these to the coefficients of the *full* Fourier transform, we need to halve the value for each nonzero n , because they are now counted twice. This distinction between the two types of transform is super important; be clear and careful about which one you're using, because otherwise you'll end up comparing apples to oranges.

In 1-dimension, and for representing real functions, we never have to consider negative wavenumbers (or, equivalently, negative n). Essentially, the reality condition implies that half of Fourier space is trivially related to the other half, so we can just pick one half and forget about the other. In two- or three-dimensions, an equivalent statement is that we only need half of the k_x, k_y plane or half of the k_x, k_y, k_z space. The reality condition in 3-d, for example, would read

$$\mathcal{A}_n(-n, -m, -l) = \mathcal{A}^*(n, m, l),$$

where n, m, l are the integer indices of the x, y, z wavenumbers. For one of the 3 directions, we need only consider positive values; however for the other two, both positive and negative wavenumbers are required. Which half we “throw out” is an arbitrary choice. But in two of the three directions, we will have positive and negative wavenumbers.

You can now attempt Exercises 2.2 and 2.3.

the Fast Fourier Transform

The slow Fourier transform approach has the advantage of being fairly easy to understand. For a Gaussian, we could derive or look up analytic expressions without too much difficulty. More generally, of course, given some initial signal, we would calculate the Fourier coefficients numerically. The fast Fourier transform is a computationally efficient way of doing this. It works by assuming a regular grid, and making use of various symmetries to speed up the numerical integration. Below is a quick introduction to python's `scipy.fft`. We won't worry about the details of how it works, but don't want to use it as too much of a black box either. To this end, we will first use it to transform some very simple functions. This will make clear how the complex Fourier coefficients correspond to the $\mathcal{A}_n$ and B_n of our “slow Fourier transform”. We will then use `scipy.fft` to consider propagation of a wave packet.

In 1-d, and using an `rfft` (which assumes the function being transformed to be real), this is straightforward. The main things to remember is that it is helpful from a numerics perspective if the number of physical grid points are easily factorable. Typically, we choose a power of 2; however, a power of 2 multiplied by some low prime is also useful. For example, if a number of grid points intermediate to $2^9 = 512$ and $2^{10} = 1024$ is desired, one might choose $3 * 256 = 768$. Another thing to keep in mind is that if the grid has values at, say, 512 points, then its Fourier transform will have 512 real numbers. These work out to be $\mathcal{A}_0$ which is real and only counts as one of the 512 needed, $\mathcal{A}_n$, $n = 1, 2, \dots, 255$ for a total of 510 real numbers. The remaining number is for the Nyquist (largest) wavenumber. This, again, is real (or sometimes pure imaginary, depending on grid choices and the specifics of the package being used). To see why this makes sense, assume the domain to go from $x = 0$

to $x = L$. But periodicity implies that $f(0) = f(L)$. So, if we define $\Delta x = L/N$, then the x positions of our grid are $0, \Delta x, 2\Delta x, \dots, L - \Delta x$. A cosine on this grid has values of $1, -1, 1, -1, \dots$ and a sine has values of zero at each point on our grid. In other words, we keep the cosine and throw out the sine.

Ordering of wavenumbers in the full fft

One thing to keep in mind with an FFT is that it returns values in an order that is counter-intuitive. Below is an attempt to explain why. This can be skipped (or merely skimmed over) if you don't really care. The main thing is to understand the wavenumbers of that the Fourier transform returns, and how to relate these to physical wavenumbers. For plotting purposes, it is often convenient to shift the order in which the wavenumbers (or FFT itself) is given, and how to do this is illustrated below.

Another point is that a Fourier transform will return complex numbers even if, as is the case below, the function being transformed is real. The real parts correspond to the A_n and the imaginary parts to the B_n . That is, the complex Fourier coefficient $\mathcal{A}_n = A_n - iB_n$ (note that i is denoted j in python). The reason that the answer comes back in what I consider to be a non-intuitive order is that, in doing the calculation, rather than manipulating integer wavenumbers going from $-N$ to N (as one might expect) one instead manipulates wavenumbers between 0 to $2N$. Here, N is the (integer) Nyquist wavenumber. This might seem odd since the Nyquist wavenumber corresponds to the smallest resolved wavelength ($2\Delta x$ in physical space), and $k = 2N$ corresponds to a wavelength on only one Δx , which is clearly not resolved by the physical grid.

To see how this makes sense, imagine that we have 8 grid points evenly spaced in x : $x = (0, 1, \dots, 7)\Delta x$, where $\Delta x = 2\pi/8$. Since we have 8 grid points, we can represent four wavelengths: L/n , $n = 1 - 4$. Suppose that for a moment we nonetheless consider $n_8 = 8$. Then $\cos(n_8 x) = 1$ and $\sin(n_8 x) = 0$ at each of our grid points.

We can then use trigonometric identities to get that

$$\begin{aligned}\cos(n_7 x) &= \cos(n_8 x - n_1 x) \\ &= \cos(n_8 x) \cos(n_1 x) + \sin(n_8 x) \sin(n_1 x) \\ &= \cos(n_1 x) = \cos(-n_1 x)\end{aligned}\tag{5}$$

Similarly, $\sin(n_7 x) = \sin(-n_1 x)$. Equivalently, $e^{in_7 x} = e^{-in_1 x}$. So, we can use the integer wavenumber ($n = 5, 6, 7$) Δk to represent the physical wavenumbers ($n = -3, -2, -1$) Δk . Recall that, if the physical space function is real, the $\mathcal{A}(k_{-n}) = \mathcal{A}(k_n)$. You can verify that this "reality condition" is indeed met in the example(s) considered below.

To summarize, the $n = 0$ wavenumber is simply the mean of the real field of which we are taking the transform. If the input is a real function, B_0 is irrelevant since $\sin(0) = 0$. Wavenumbers $n = 1, 2, 3$ are as you would expect; they correspond to 1, 2, or 3 wavelengths fitting in the domain. $n = 4$ is special in that $k_4 x$ is an integer multiple of π for each value on our grid. So, $\cos(k_4 x) = \pm 1$ at each of the x on our regular grid. Conversely, $\sin(k_4 x)$ is zero everywhere. Because of this, in the Fourier transforms below, there is a

value corresponding to $k = -k_4$ but not for $k = +k_4$. Because of the symmetry (i.e., because the sine is zero anyway), less coefficients are needed, and so less are given (with the total number of complex coefficients corresponding to half the total number of physical grid points).

Finally, n_5 corresponds to n_{-3} , n_6 to n_{-2} , and n_7 to n_{-1} . When reporting the k values, the FFT will give $(0, 1, 2, 3, -4, -3, -2, -1)$ times some Δk . The python function `fftshift` puts the results back in a somewhat more sensible order, $(-4, -3, -2, -1, 0, 1, 2, 3)\Delta k$. Here, $\Delta k = 2\pi/L$, in which L gives the assumed periodicity of the domain.

Again, it is worth reading through this and playing around with different choices for $f(x)$ to get a feel for how the notation all works. Also, you may later use a different FFT (perhaps in a different programming language). It is always a good idea to first test it out with simple functions to make sure you are interpreting the results correctly (e.g., the results might be normalized in some way that you did not anticipate. (An alternative, of course, would be to read the documentation...))

Now try Exercise 2.4, on FFTs!

Chapter 3: Dispersion

Given Fourier series representations of f_1 and f_2 , it becomes clear that each Fourier component propagates either to the left or to the right at the same speed, c . In other words if we think of ct as the phase of the sinusoids comprising f_1 and $-ct$ as the phase of those comprising f_2 , then since the phases of all the sine waves comprising each of f_1 and f_2 are the same at any instant in time, summing them together produces the same signal. Simply, in the case of f_1 this signal has been displaced to the left, and for the case of f_2 , to the right. These are called nondispersive waves.

Dispersion occurs when different Fourier components travel at different speeds. Before we go any further, attempt Exercise 3.1 to derive some basic relations in the case of *wavepackets* that we will need to treat the problem of dispersion.

In 2- or 3-dimensions, it can also occur with different wavelengths propagate in different directions. Another example of dispersive waves is that of surface gravity waves, such as those that might make you seasick in a small boat out in rough seas. We will not discuss these waves in detail but will try to infer how ω depends on k for these waves using something called dimensional analysis. Dimensional analysis is a cornerstone of turbulence theory. The idea in using it here is just to introduce the basic idea (as a sort of aside to the main subject of this module) and then to derive without doing too much work the dispersion relation, $\omega(k)$ for a special case of surface gravity waves —those in deep water. By deep water, we mean that the water depth is sufficiently deep that the waves don't interact with the bottom. Typically, a depth greater than about half the wavelength is deep enough.

Deep-water gravity waves also provide a nice starting point since they are familiar to anyone who has been to the beach or watched the waves from a boat wake as they approach the shore. Fixing your eye on a particular wavecrest in the packet of waves making up a boat wake, you may have noticed that the crest moves forward through the packet with

time. That is, while the entire packet is moving forward, individual crests are also moving forward, but at a faster speed. As such, fixing your eye on the crest or a small wave that is initially near the back of the packet, you will first see it grow in amplitude and move towards the center of the pack. Later, it will shrink in amplitude as it moves towards the front of the packet, before eventually disappearing. You may also have noticed (sitting on a dock, say, as a boat goes by somewhere offshore) that the first waves to arrive at the dock have relatively low frequencies and long wavelengths. As time passes, the waves arriving have successively shorter wavelengths and higher frequencies. This, again, is dispersion. Dispersion occurs when the phase speed (or more generally, the phase velocity) is different from the group speed (or group velocity). The phase speed is simply the speed at which a particular Fourier component of a signal propagates. In one dimension, it is given simply by $c_{\text{phase}} = \omega/k$. The group speed is the speed at which the interference pattern formed by summing Fourier components with similar values of k propagates.

Dispersion relation for deep-water surface gravity waves

To use dimensional analysis, let us assume that the frequency depends on gravity ($g = 9.8\text{ms}^{-2}$) and on the wavelength, λ . Equivalently, it depends on g and the wavenumber, $k = 2\pi/\lambda$. Note that small wavenumbers correspond to large wavelengths (and vice versa). What about other parameters? For linear waves, we don't expect a dependence on the amplitude. For example, a note played softly on a guitar has the same frequency as the same note played louder. Similarly, if the depth of the water is sufficiently deep, the actual depth H is effectively infinite and thus irrelevant. Other factors might also matter. Surface tension, for example, affects the wave properties, but this is important only for very short wavelengths (e.g., ripples, or capillary waves). Similarly, one might anticipate that the wave speed depends on density; however, since the density of water is large compared to that of the overlying air, the exact value turns out to be irrelevant. We thus exclude these factors from consideration.

We want an expression for ω and postulate that it must depend on g and k : $\omega = \omega(g, k)$. Let's look at the units:

$$\omega \sim \text{s}^{-1} \quad ; \quad k \sim \text{m}^{-1} \quad ; \quad g \sim \text{ms}^{-2}$$

Functions such as sine, cosine, exponentials and such have nondimensional arguments (as is evident, for example, from a Taylor's Series expansion). This observation leads to $\omega = g^a k^b$, where a and b are nondimensional exponents and are to be determined. We determine them by getting the dimensions right.

On the left hand side, we have that ω goes like s^{-1} . The only time on the right hand side is in g ; hence, $a = 1/2$. Then, $\omega/g^a \sim \text{m}^{-1/2}$, which must match with k^b , implying $b = 1/2$. We find from this that $\omega \propto (gk)^{1/2}$. This dimensional analysis does not give us the proportionality constant. But if we were to measure it for one k , it would work for all. Similarly, if we could somehow change g (go to another planet, say), the same proportionality constant would hold. Constants such as this are called "universal constants". Somewhat

fortuitously, in this case the proportionality constant works out to be unity. That is,

$$\omega = (gk)^{1/2}.$$

A more complete treatment of *general* surface gravity waves, which may interact with the bottom and thus care about the total depth, gives that

$$\omega^2 = gk \tanh(kH),$$

where H is water depth. Note that, for long waves $kH \ll 1$, this reduces to $\omega^2 \approx c^2 k^2$, where $c^2 = gH$. That is, we recover nondispersive waves travelling at speed $(gH)^{1/2}$. An example might be a tsunami travelling across the ocean. For deep water waves, $kH \gg 1$, the hyperbolic tangent asymptotes to one, and we recover $\omega^2 = gk$, consistent with what we found above using dimensional analysis. In this deep water case, $(\omega/k) = c = (g/k)^{1/2} = \alpha \lambda^{1/2}$, where the constant $\alpha = (g/2\pi)^{1/2}$. In other words, c in ms^{-1} is about 1.25 times the square root of the wavelength in meters.

Now attempt the rest of exercise 3. This is harder and requires more thought, interpretation and choices about what to explore than the previous exercises, so take your time: especially with 3.2-3.3